

CICD REPORT

1. CI/CD Pipeline Configuration

The project uses GitHub Actions to automatically execute API tests using Newman. The pipeline is triggered whenever code is pushed to or a pull request is created for the master branch. It can also be started manually using workflow_dispatch.

The workflow file is: .github/workflows/api-tests.yml

The pipeline consists of the following steps:

1. **Checkout repository** – retrieves the project source code.
2. **Setup Node.js** – configures Node.js 24 on the GitHub Actions runner.
3. **Install backend dependencies** – runs npm install in eshop-sut/backend.
4. **Start backend** – starts the EShop backend before API testing.
5. **Wait for backend** – verifies that the API is available on localhost:3000.
6. **Install Newman** – installs Newman for executing Postman collections.
7. **Install HTML Extra Reporter** – installs newman-reporter-htmlextra for generating HTML reports.
8. **Run API tests** – executes the Postman collection against the EShop API.
9. **Upload Newman reports** – stores the generated HTML reports as GitHub Actions artifacts.

2. Successful Pipeline Run

For the successful run, all Newman assertions passed and the GitHub Actions job completed successfully.

Metric	Result
Requests	35
Assertions	35
Failed Requests	0
Passed Requests	35

Action Run Link: https://github.com/lebao05/Testing_Homework6/actions/runs/33236988020

Summary

All jobs

api-tests

Run details

Usage

Workflow file

api-tests

succeeded 7 minutes ago in 24s

Run FR10 - Cancel Order

```

180 PUT http://localhost:3000/api/orders/134023524/cancel [404 Not Found, 301B, 1ms]
181 ✓ Status code is expected
182 • TC-CANCEL-011 Success Response Schema
183 PUT http://localhost:3000/api/orders/14/cancel [200 OK, 308B, 3ms]
184 ✓ Status code is expected
185 • TC-CANCEL-012 Error Response Schema
186 PUT http://localhost:3000/api/orders/3/cancel [400 Bad Request, 313B, 1ms]
187 ✓ Status code is expected
188 • TC-CANCEL-013 After Refund Processed
189 PUT http://localhost:3000/api/orders/18/cancel [400 Bad Request, 313B, 3ms]
190 ✓ Status code is expected
191 • TC-CANCEL-014 Missing Content-Type
192 PUT http://localhost:3000/api/orders/9/cancel [200 OK, 308B, 3ms]
193 ✓ Status code is expected
194 • TC-CANCEL-015 verify status changed
195 PUT http://localhost:3000/api/orders/15/cancel [200 OK, 308B, 3ms]
196 ✓ Status code is expected

```

	executed	failed
iterations	1	0
requests	35	0
test-scripts	35	0
prerequisite-scripts	0	0
assertions	35	0

```

197 total run duration: 540ms
198
199 total data received: 2.5KB (approx)
200
201 average response time: 3ms [min: 1ms, max: 40ms, s.d.: 7ms]

```

Upload Newman Reports

```

1 Node 20 is being deprecated. This workflow is running with Node 20 by default. If you need to temporarily use Node 20, you can set the ACTIONS_ALLOW_USE_UNSECURE_NODE_VERSION=true environment variable. For more information see:
https://github.blog/changelog/2025-09-18-deprecation-of-node-20-on-github-actions-runners/
2 Run actions/upload-artifact@v4
3 (node:2048) [DEP0040] deprecationWarning: The "punycode" module is deprecated. Please use a userland alternative instead.
4 (Use "node --trace-deprecation ..." to show where the warning was created)

```

Successful Run

github.com/lebaos05/testing_Homework6/actions/runs/33236988020

Tài liệu sinh viên - G... Khoa Công nghệ Th... CS229 Machine Lea... CS230 Deep Learning The essence of calcu... Khôn GPA Portal v2.0 Practical Statistics fo... Essential Grammar ... English_Grammar_in... Cambridge-English... 1_English_Vocabular... 2_English_Vocabular...

lebaos05 / Testing_Homework6

Type to search

Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

API Tests - Newman

Update shop #5

Re run all jobs

Summary

All jobs

api-tests

Run details

Usage

Workflow file

Triggered via push 8 minutes ago

lebaos05 pushed • [@lebaos05](#) [master](#)

Success

Total duration 27s

Artifacts 1

api-tests.yml

api: push

api-tests

24s

Annotations

1 warning

api-tests

Node.js 20 is deprecated. The following actions target node.js 20 but are being forced to run on node.js 20: actions/checkout@v4, actions/setup-node@v4, actions/upload-artifact@v4. For more information see: <https://github.blog/changelog/2025-09-18-deprecation-of-node-20-on-github-actions-runners/>

Show more

Artifacts

Produced during runtime

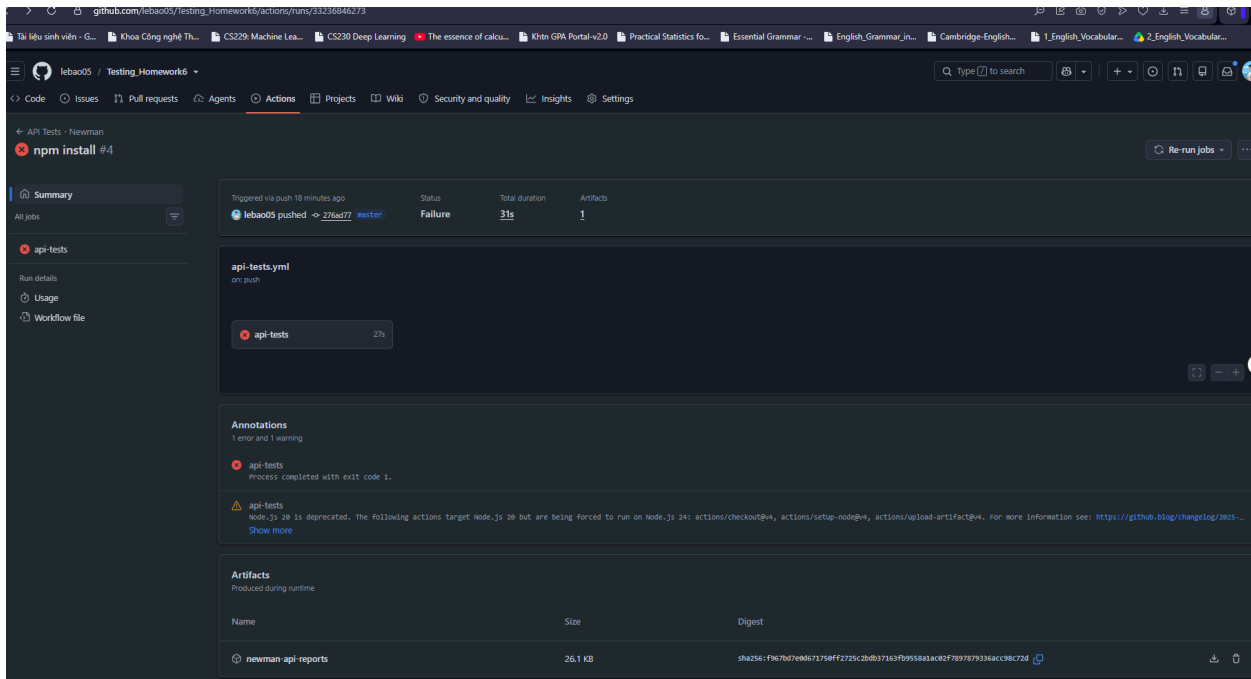
Name	Size	Digest
newman-api-reports	25.6 KB	sha256:ef88e192c565e9adcb662963f881dab7721d1385689676478782412705f658

Successful Run

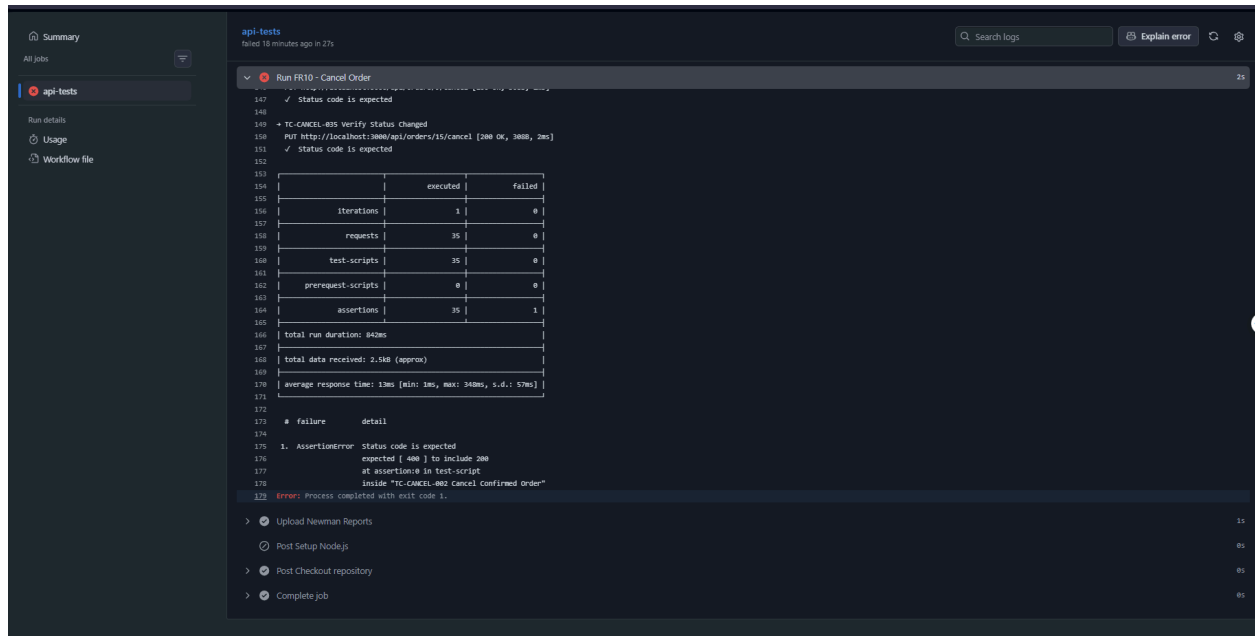
3. Failing Pipeline Run

Metric	Result
Requests	35
Assertions	35
Failed Requests	1
Passed Requests	34
Failed Test Case	TC-CANCEL-002

Action Run Link: https://github.com/lebao05/Testing_Homework6/actions/runs/33236846273



Failed Run



Failed Run

4. Conclusion

The implemented CI/CD pipeline integrates GitHub Actions, Newman, and Postman to automate API testing for the EShop system.

The successful pipeline demonstrates that the system can automatically execute the API tests and generate test reports when all assertions pass. The failing pipeline demonstrates that an incorrect API response is detected by Newman and causes the GitHub Actions job to fail.

Therefore, the pipeline provides an automated mechanism for detecting API test failures during the development and integration process.